Dual-Device System Architecture Flowchart

Dual-Viewport Inspection (Mobile 390px + Desktop 1440px), 7-Stream Concurrency & Device-Aware Scoring

Architecture: Dual-Device Local-First | Viewports: Mobile 390px & Desktop 1440px | Diagnostic Streams: 7 Parallel Workers across 2 Phases | Dual Vitals: PageSpeed v5 Mobile + Desktop | Scoring: Separate Mobile & Desktop Health

1 STEP 1: PRESENTATION TIER & DEVICE FILTER INTERFACE

□ React 18 + Vite Dashboard with Device Filter System (frontend/src/App.tsx)

Device: All | Mobile | Desktop

- User inputs target website URL and clicks 'Run Diagnostic Audit'. Features real-time Device Filtering ('all', 'mobile', 'desktop').
- Establishes reactive HTTP EventSource connection to SSE endpoint: /api/audit/stream?url=<target>.
- Tracks 7 active workers concurrently: network, dom_seo, saudi, mobile (390px), desktop (1440px), external (vitals), aeo_geo.

HTTP GET /api/audit/stream (StreamingResponse)

2 STEP 2: GATEWAY & DUAL-DEVICE ASYNC LAUNCH

□ FastAPI ASGI Core & scan_headless_devices Orchestrator (backend/main.py • Uvicorn :8000)

Concurrent Browser Contexts

- Generates audit_id UUID. Executes pre-flight raw HTML fetch via httpx.AsyncClient with desktop User-Agent.
- Dispatches Phase 1 tasks: scan_headless_devices (spawns mobile 390px + desktop 1440px), network_scanner, and dual PageSpeed.
- Streams atomic milestone updates ('worker_start': mobile, desktop, network, external) to client EventSource.

Phase 1 Launch: scan_headless_devices + network + external intel

3 STEP 3: THE 7 DIAGNOSTIC WORKER STREAMS (TWO-PHASE MATRIX)

Dual Viewports & Dual Vitals

Worker 1: Network & SSL — Phase 1

network_scanner.py

- TLS 1.2/1.3 socket handshake
- SSL expiry countdown days
- HTTP □ HTTPS 301/308 redirect
- TTFB latency rating (ms)
- Server header & CDN detection

Worker 4: Playwright Mobile — Mobile UX

mobile_browser.py (390px)

- iPhone 14 Pro emulation (390px)
- Tap targets < 48px touch audit
- Mobile scroll overflow check
- Captures mobile fold & full PNG
- Local FCP & load time metrics

Worker 5: Playwright Desktop — Desktop UX

mobile_browser.py (1440px)

- Desktop Viewport (1440x900px)
- Desktop horizontal layout overflow
- Captures desktop fold & full PNG
- Extracts rendered dynamic DOM
- Identifies overflow DOM elements

Worker 6: Dual PageSpeed — Dual Vitals

external_intel.py (v5 API)

- Mobile Lighthouse Scores (0-100)
- Desktop Lighthouse Scores (0-100)
- Core Web Vitals: LCP, CLS, INP
- Speed Index & Total Blocking Time
- Next-Gen image format savings

□ **Phase 2 Handoff:** Playwright yields rendered dynamic DOM (mobile or desktop) □ passed concurrently to Workers 2, 3, and 7

Hydrated DOM Feed

Worker 2: DOM & Technical SEO — Technical SEO

dom_seo_scanner.py • BeautifulSoup4

- Title tag length (30-60 chars) & Meta Description
- Single H1 semantic heading hierarchy validation
- Image alt text accessibility ratio & total count
- robots.txt rules & sitemap.xml presence check
- Pixels: GA4, GTM, Meta, Snapchat, TikTok

Worker 3: Saudi Conversion — GCC Conversion

saudi_scanner.py • Localization RegEx

- Direct WhatsApp link detection (wa.me)
- Click-to-call mobile phone tags (tel:)
- Saudi Gateways: Mada, Apple Pay, STC Pay
- Buy-Now-Pay-Later gateways: Tabby & Tamara
- Arabic RTL direction & Lorem ipsum placeholders

Worker 7: Generative AEO & GEO — AEO / AI Search

aeo_scanner.py • Schema.org & LLM SEO

- AI crawler permissions: GPTBot, PerplexityBot
- ClaudeBot & Google-Extended robots directives
- Schema.org JSON-LD structured data parser
- sameAs Knowledge Graph entity authority links
- RAG direct-answer chunkability & FAQ structures

Aggregate Device-Specific Issue Sets, Scores & Dual PNG Artifacts

4 STEP 4: DUAL-DEVICE SCORING & MULTIMODAL GEMINI AI

□ Dual-Device Scoring & Metrics Evaluator (services/scoring_engine.py)

- Calculates overall_score, mobile_overall_score, and desktop_overall_score.
- Device-Aware Filtering: Desktop score skips Mobile UX tap targets; Mobile score skips Desktop UX overflow.
- Evaluates 20+ checks tagged with device='all', 'mobile', or 'desktop' against W3C benchmarks.
- Synthesizes actionable remediation explanations and copy-paste code snippets for engineers.

□ Google Gemini Multimodal Sales Pitch Engine (services/ai_pitch_engine.py)

- Multimodal Vision: Inspects captured mobile or desktop fold screenshot for visual hierarchy and CTA clarity.
- Model Fallbacks: Tries gemini-2.0-flash, gemini-1.5-flash, gemini-2.5-flash for personalized outreach copy.
- Deterministic Baseline: Instant fallback to high-converting Arabic WhatsApp scripts and bilingual cold emails.
- Discovery Questions: 4 qualification questions uncovering bottlenecks in client lead routing.

Save Dual Scores to SQLite & Dispatch SSE 'event: complete'

5 STEP 5: PERSISTENCE & DEVICE-FILTERABLE DASHBOARD DELIVERY

□ Local SQLite & Dual Screenshots Storage (audits.db • /screenshots/)

- audits table (aiosqlite): Stores audit UUID, target URL, overall/mobile/desktop scores, and full JSON.
- leads table: Built-in CRM tracking prospective clients, outreach status, decision-makers, and notes.
- Dual Screenshots: Saves both {id}_mobile_fold.png and {id}_desktop_fold.png for interactive review.

□ Interactive Presentation & Viewport Switcher (event: complete received)

- Device Filter Tabs: Toggle checklist and analytics between 'All', 'Mobile', and 'Desktop'.
- Dual Scorecards: Side-by-side radar scorecards and progress gauges for mobile vs desktop.
- Dual Viewport Mock: Switch between iPhone 14 Pro mock and Desktop 1440px viewport.
- AI Outreach Center: One-click copy for Arabic WhatsApp script and bilingual sales pitches.

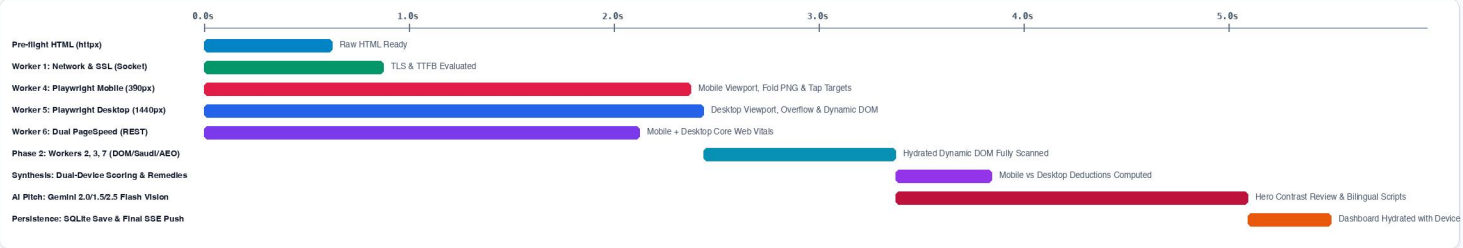
Diagnostic Workers Specification Matrix (Dual-Device)

Detailed technical breakdown of inputs, diagnostic routines, target benchmarks, and latency budgets

1. THE 7 DIAGNOSTIC WORKER STREAMS SPECIFICATION TABLE

#	Worker & Module	Engine / Library	Targeted Inspections & Attributes	Benchmark Standards	Device	Phase
01	**Worker 1: Network & Security** workers/network_scanner.py	socket, ssl, httpx	Direct TLS 1.2/1.3 socket handshake on port 443; SSL expiration days countdown; CA issuer chain; HTTP → HTTPS redirect codes (301/308); TTFB latency; CDN header identification.	SSL > 30 days HTTP → HTTPS: 301 TTFB < 300ms	**All Devices** (General)	**Phase 1** (Initial)
02	**Worker 2: DOM & SEO Scanner** workers/dom_seo_scanner.py	BeautifulSoup4, re	Title length & meta description presence; single H1 hierarchy; image alt text coverage ratio; robots.txt disallow rules & sitemap.xml presence; GA4, GTM, Meta Pixel, Snap & TikTok tracking.	Title: 30–60 chars H1 count == 1 Alt coverage: 100%	**All Devices** (General)	**Phase 2** (Hydrated)
03	**Worker 3: Saudi Conversion** workers/saudi_scanner.py	Localization RegEx	Direct WhatsApp lead channels (wa.me); click-to-call phone links (tel:); Saudi payment gateways (Mada, Apple Pay, STC Pay, Tabby, Tamara); Arabic RTL orientation; placeholder text.	WhatsApp & tel: present Mada & Apple Pay enabled RTL dir attribute	**All Devices** (General)	**Phase 2** (Hydrated)
04	**Worker 4: Playwright Mobile** workers/mobile_browser.py	Playwright (390px)	Emulates iPhone 14 Pro (390×844, DPR 3.0); measures layout overflow (scrollWidth > clientWidth); calculates tap target bounding boxes (< 48px); captures fold and full screenshots.	scrollWidth == clientWidth Tap targets >= 48px Hero fold capture	**Mobile** (Viewport)	**Phase 1** (Initial)
05	**Worker 5: Playwright Desktop** workers/mobile_browser.py	Playwright (1440px)	Emulates Desktop Viewport (1440×900); measures desktop layout horizontal overflow; captures desktop fold and full screenshots; extracts rendered dynamic SPA DOM for Phase 2.	scrollWidth == 1440px Desktop fold capture Dynamic DOM yield	**Desktop** (Viewport)	**Phase 1** (Initial)
06	**Worker 6: Dual PageSpeed** workers/external_intel.py	Google Cloud REST (v5)	Queries Google PageSpeed Insights REST API for both Mobile and Desktop Lighthouse audits: FCP, Speed Index, TBT, LCP, CLS, INP, and next-gen image compression savings.	LCP < 2.5s, CLS < 0.1 INP < 200ms Desktop Score >= 90	**Mobile &** Desktop	**Phase 1** (Initial)
07	**Worker 7: Generative AEO & GEO** workers/aeo_scanner.py	Schema.org, JSON-LD	AI search crawler permissions in robots.txt (GPTBot, PerplexityBot, ClaudeBot, Google-Extended); Schema.org JSON-LD structured data; sameAs authority links; RAG direct-answer chunkability.	AI Bots: Allowed sameAs links > 0 FAQ schema present	**All Devices** (General)	**Phase 2** (Hydrated)

2. DUAL-DEVICE ASYNCHRONOUS TWO-PHASE TIMELINE (GANTT PROFILE)



Data Architecture, Domain Models & Deployment

SQLite database schemas, Pydantic domain models, security profile, and local deployment blueprint

1. ASYNCHRONOUS SQLITE DATABASE ARCHITECTURE (audits.db)

TABLE: audits (Dual-Device JSON & Scores)

Stores all synthesized dual-device audit payloads for instant recall

```
CREATE TABLE IF NOT EXISTS audits (
    id TEXT PRIMARY KEY,               -- UUID v4
    target_url TEXT NOT NULL,          -- Normalized Target Domain
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    overall_score INTEGER NOT NULL,    -- 0-100 General Health
    data_json TEXT NOT NULL            -- Serialized Dual Payload
);
```

TABLE: leads (Built-In Sales CRM Pipeline)

Tracks prospective client outreach status, decision-makers, and notes

```
CREATE TABLE IF NOT EXISTS leads (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    audit_id TEXT REFERENCES audits(id),
    company_name TEXT, contact_person TEXT,
    phone TEXT, email TEXT,
    status TEXT DEFAULT 'new',         -- new, contacted, closed
    notes TEXT, updated_at TIMESTAMP
);
```

2. PYDANTIC DOMAIN MODEL ARCHITECTURE (models.py)

FullAuditResponse

Root Synthetic Model

- audit_id: str (UUID)
- target_url: str
- overall_score: int
- mobile_overall_score: int
- desktop_overall_score: int
- mobile: HeadlessMobileResult
- desktop: HeadlessDesktopResult
- evaluated_metrics: List

MetricEvaluation

Device-Aware Check

- id: str (rule identifier)
- name: str (human title)
- category: str
- status: MetricStatus
- score: int (0 to 100)
- device: 'all'/'mob'/'desk'
- benchmark: str
- code_snippet: Optional[str]

HeadlessDesktopResult

Playwright Desktop

- horizontal_overflow: bool
- viewport_width: 1440
- viewport_height: 900
- scroll_width: int
- overflow_elements: list
- fold_screenshot_path: str
- full_screenshot_path: str
- rendered_html: str

ExternalIntelResult

Dual PageSpeed Vitals

- mobile_performance_score
- desktop_performance_score
- lcp_seconds (mobile)
- desktop_lcp_seconds
- cls_score & desktop_cls
- inp_ms & desktop_inp
- image_savings_mb
- issues: List[AuditIssue]

3. OPERATIONAL PROFILE & PRODUCTION DEPLOYMENT BLUEPRINT

□ One-Command Local Startup

Execute from the project repository root:

```
#!/usr/bin/env bash
./run.sh

# Automatically executes:
# 1. Installs/verifies virtualenv dependencies
# 2. Builds React 18 production bundle (Vite)
# 3. Initializes local SQLite database
# 4. Launches FastAPI on http://127.0.0.1:8000
```

□ Engineering Invariants & Security Guardrails

- Dual-Device Isolation: Playwright launches isolated mobile (390px) & desktop (1440px) contexts cleanly.
- Zero Infrastructure Cost: $0.00/month recurring cost. Runs completely on developer localhost.
- Complete Data Privacy: Target audit records and screenshot files remain local in audits.db and /screenshots.
- Graceful AI Multi-Model Fallbacks: Tries gemini-2.0-flash, 1.5-flash, 2.5-flash with local copy fallbacks.
- Network Timeout Controls: Strict 10-second request timeouts protect against slow, unresponsive targets.